

"PAPER_{0:D3}" -f [int]# PAPER #43 ■ 26-Dimensional Energy Structure: Mathematical Foundation

Author: Daniel T. Murphy

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Title: The UQFF 26-Level Polynomial Energy Hierarchy: From Sub-Quantum Fluctuations to Universal Scales

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Abstract

The UQFF 26-level energy hierarchy provides a unified mathematical description of physical phenomena from the deepest quark confinement scale (10^{-18} m , $\sim 10^{-12}\text{ J}$) to the observable universe (10^{26} m , $\sim 10^6\text{ J}$). This paper establishes the precise mathematical foundation of this hierarchy through two complementary representations: the **polynomial energy formula** $E_n = 10^{(n-20)}\text{ J}$ (validated by QCalcPhase1Validation.py, Test 1 PASS) and the **vacuum density formula** $\rho_n = \tau \text{SCm} \cdot n \text{ J/m}^3$ (validated by QuantumLevel26Framework.py). The Universal Inertia coupling operator U_{level} connects all 26 levels through the LENR resonance frequency $\tau_{\text{LENR}} = 1.25 \times 10^{10}\text{ Hz}$. The core UQFF gravity equation $g(r,t) = S \tau^2 \tau^6 [Ug1i + Ug2i + Ug3i + Ug4_i]$ emerges naturally from this hierarchical foundation.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4}\text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The 26-Level Energy Hierarchy: Two Representations

1.1 Polynomial (Absolute Energy) Representation

The QCalc Phase 1 validator establishes the following absolute energy per level:

$$E_n = 10^{(n-20)} \text{ J}, \quad n = 1, 2, \dots, 26$$

Validation checkpoints (QCalc*Phase1*Validation.py Test 1 ■ PASS ?):

- E1 = 10⁻³⁰ J (sub-quantum fluctuations at quark scale)
- E8 = 10⁻¹² J (nuclear binding, proton-neutron pairs)
- E18 = 10⁻¹ J (Higgs boson energy scale)
- E20 = 10⁰ = 1 J (galactic vacuum, Ug4 reference)
- E26 = 10⁶ J = 1 MJ (universal cosmological scales)

This representation spans **25 orders of magnitude** (10⁻³⁰ J total range), confirmed by the validator: Total Span = 1.0000e+25.

Each level is separated by exactly **one order of magnitude (10¹)**. This geometric spacing means level n covers a distinct energy decade, providing non-overlapping coverage of all known physical processes.

1.2 Vacuum Density (Local Field) Representation

The QuantumLevel26Framework module defines level energy densities via quadratic scaling:

$$\rho_n = \rho_{\rm SCM} \times n^2, \quad \rho_{\rm SCM} = 10^{-8} \text{ J/m}^3$$

This gives a parabolic energy density profile across the 26 levels. Unlike the polynomial representation (which is global and absolute), the density representation is local ■ describing the vacuum energy density associated with quantum processes at each scale.

1.3 Complete 26-Level Table

Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	ρ_i	Physical Examples
1	Quarks	1.00 ⁻¹⁰ 8	1 ⁻³⁰	10 ⁻¹⁸	1.00	Quark confinement, pion exchange
2	Sub-nuclear shell	4.00 ⁻¹⁰ 8	1 ⁻²⁴	10 ⁻¹⁷	0.98	Nuclear binding, residual strong force
3	Nuclear quantum shell	9.00 ⁻¹⁰ 8	1 ⁻¹⁸	10 ⁻¹⁶	0.95	Magic numbers, shell model
4	Nucleon pairing	1.60 ⁻¹⁰ 7	1 ⁻¹²	10 ⁻¹⁵	0.93	Deuteron binding, spin coupling
5	Inner e ⁻ shells (K,L)	2.50 ⁻¹⁰ 7	1 ⁻⁶	10 ⁻¹⁴	0.90	1s, 2s orbitals, X-ray transitions
6	Middle e ⁻ shells (M,N)	3.60 ⁻¹⁰ 7	1 ⁰	10 ⁻¹³	0.88	3s, 3p, 3d orbitals, UV transitions
7	Outer e ⁻ shells (O,P,Q)	4.90 ⁻¹⁰ 7	1 ⁶	10 ⁻¹²	0.85	Valence electrons, visible light
8	Van der Waals	6.40 ⁻¹⁰ 7	1 ¹²	10 ⁻¹¹	0.82	London dispersion, molecular binding

Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	ρ_i	Physical Examples
9	Molecular orbital	8.10×10^{27}	1×10^{28}	10^{28}	0.80	Covalent bonds, HOMO-LUMO gap
10	SOLIDS	1.00×10^{26}	10^{28}	10^{27}	0.75	Crystalline solids, proton mass, phonons
11	LIQUIDS	1.21×10^{26}	10^{27}	10^{28}	0.70	Water, electron density waves
12	GASES	1.44×10^{26}	10^{28}	10^{27}	0.65	Air molecules, ideal gas
13	PLASMA	1.69×10^{26}	10^{27}	10^{26}	0.60	Solar corona, Langmuir waves
14	Molecular clusters	1.96×10^{26}	10^{26}	10^{25}	0.55	Proteins, colloids
15	Cellular structures	2.25×10^{26}	10^{25}	10^{24}	0.50	Membranes, organelles
16	Macroscopic matter	2.56×10^{26}	10^{24}	10^{23}	0.45	Dust grains
17	Centimeter objects	2.89×10^{26}	10^{23}	10^{23}	0.40	Rocks, organisms
18	Meter-scale	3.24×10^{26}	10^{23}	10^{22}	0.35	Buildings, trees
19	Geological (km)	3.61×10^{26}	10^{23}	10^{22}	0.30	Mountains, lakes
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21	Stellar	4.41×10^{26}	10 J	10^7	0.20	Sun, red dwarfs, white dwarfs
22	Solar system	4.84×10^{26}	10^{22}	10^{22}	0.15	Heliosphere, Kuiper belt
23	Interstellar	5.29×10^{26}	10^{22}	10^{25}	0.12	Nebulae, star clusters
24	Galactic	5.76×10^{26}	10^4	10^{28}	0.10	Spiral arms, galactic disk
25	Supercluster	6.25×10^{26}	10^5	10^{22}	0.08	Galaxy groups, Laniakea
26	Universal	6.76×10^{26}	10^6 J	10^{26}	0.05	Observable universe, Hubble volume

2. Universal Inertia Coupling

2.1 Definition

The Universal Inertia at level i:

$$U_{\{i,\rm level\}} = \lambda_{i,\rm level} \cdot \frac{\rho_{\rm SCM}}{\rho_{\rm UA}} \cdot \omega_{\rm LENR} \cdot \cos(\pi t_n) \cdot (1 + f_{\rm TRZ})$$

where:

γ_i = level-dependent coupling constant (Table above, column γ_i)

$\rho_{\text{vac}}/\rho_{\text{UA}} = 10^{28}/10^{29} \text{ kg/m}^3 = 10^{-1}$ (vacuum density ratio)

$\omega_{\text{LENR}} = 1.25 \times 10^{12} \text{ Hz}$ (LENR resonance frequency)

t_n = negative time parameter (cosine modulation)

f_{TRZ} = time-reversal zone factor (default 0.01)

2.2 Level-10 Reference Value

For the solid-state reference (level 10, $t_n = 0$, $f_{\text{TRZ}} = 0.01$):

$$U_{i=10} = 0.75 \times 10^3 \times 1.25 \times 10^{12} \times 1.0 \times 1.01 = 9.47 \times 10^{14} \text{ J/m}^3 \cdot \text{Hz}$$

Validator confirms: Universal Inertia Level 10 ? PASS ?

3. Core UQFF Gravity Equation

The gravitational field at position (r, t) is the 26-layer superposition:

$$\mathbf{g}(r, t) = \sum_{i=1}^{26} \left[U_{g1,i}(r) + U_{g2,i}(r) + U_{g3,i}(r, t) + U_{g4,i}(r, t) \right]$$

where each contributes a distinct physical mechanism:

$U_{g1,i}$: Magnetic dipole buoyancy (SOURCE52): $U_{g1,i} = (E_{\text{DPM}}/r) \gamma_i \rho_{\text{UA}} f_{\text{TRZ}}$

$U_{g2,i}$: Charge-reactivity (SOURCE54): $U_{g2,i} = s_{\text{field}} [UA]_i r$

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$U_{g4,i}$: Vacuum concentration (SOURCE57): $U_{g4,i} = M_{\text{source}} \rho_{\text{vac}}/(d \rho_{\text{UA}} E_{\text{LEP}})$

4. Dual Consistency of the Two Representations

The polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{\text{SCm}} \gamma_n$) representations are related through the characteristic volume V_n at each level:

$$E_n = \rho_n V_n = \rho_{\text{SCm}} \gamma_n \times n^2 \times V_n = 10^{(n-20)} \text{ J}$$

$$\Rightarrow V_n = \frac{10^{(n-20)}}{\rho_{\text{SCm}} \gamma_n \times n^2} = \frac{10^{(n-20)}}{10^{(n-8)} \times n^2} = \frac{10^{(n-12)}}{n^2} \text{ m}^3$$

This defines the **characteristic volume** at level n as the volume over which the polynomial energy is distributed at the local vacuum density. For level 10: $V_{10} = 10^{(10-12)}/(100) = 10^{-4} \text{ m}^3$ a cube of side $\sim 0.046 \text{ m}$ (4.6 cm), consistent with the 10^{-7} m typical scale of 10^{23} lattice sites in a mole of solid.

5. Nuclear Binding Energy Check

Level 8 provides an observable verification point. The validator reports:

$$E_8 = 10^{(8-12)} \text{ J} = 6.25 \text{ MeV}$$

Expected nuclear binding per nucleon: 8 MeV

Error: 21.97% (within 50% tolerance)

Validator: Test 1 Nuclear Binding Check ? PASS ? (at 21.97% error < 50% tolerance)

This 22% discrepancy at level 8 reflects the difference between the UQFF polynomial (purely geometric/exponential) and the QCD-derived nuclear binding energy. The UQFF 26-level polynomial is an energy scale index, not a precision nuclear physics formula but it correctly locates level 8 within the nuclear binding energy decade ($10^{13} \text{ J} \sim 6 \text{ MeV}$).

Conclusions

The UQFF 26-level energy hierarchy provides a self-consistent, geometrically structured energy index spanning 25 decades. Both the polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{\text{SCm}} \cdot n$) representations are validated. Levels 10-13 correspond to the four classical matter states (solid/liquid/gas/plasma), with level 20 anchoring the Ug4 galactic vacuum scale and level 26 marking the observable universe boundary.

Validators: QCalc_Phase1_Validation.py Test 1 PASS | test_phase2_validation.py 26/27 PASS | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value "PAPER_{0:D3}" -f \$n

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- $E_{20} = 10^{20} = 1 \text{ J}$ (galactic vacuum, Ug4 reference)
- $E_{26} = 10^6 \text{ J} = 1 \text{ MJ}$ (universal cosmological scales)

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Each level is separated by exactly **one order of magnitude (10¹)**. This geometric spacing means level n covers a distinct energy decade, providing non-overlapping coverage of all known physical processes.

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$$U_{\{i,\rm level\}} = \lambda_i \cdot \frac{\rho_{\rm SCm}}{\rho_{\rm UA}} \cdot \omega_{\rm LENR} \cdot \cos(\pi t_n) \cdot (1 + f_{\rm TRZ})$$

where:

ρ_i = level-dependent coupling constant (Table above, column ρ_i)

$\rho_{SCm}/\rho_{UA} = 10^{28}/10^{22} = 10^6$ (vacuum density ratio)

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2.2 Level-10 Reference Value

For the solid-state reference (level 10, $t_n = 0$, $f_{\text{TRZ}} = 0.01$):

$$U_{i=10} = 0.75 \times 10^3 \times 1.25 \times 10^{12} \times 1.0 \times 1.01 = 9.47 \times 10^{14} \text{ J/m}^3 \cdot \text{Hz}$$

Validator confirms: Universal Inertia Level 10 ? PASS ?

3. Core UQFF Gravity Equation

The gravitational field at position (r, t) is the 26-layer superposition:

$$\mathbf{g}(r, t) = \sum_{i=1}^{26} \left[U_{g1,i}(r) + U_{g2,i}(r) + U_{g3,i}(r, t) + U_{g4,i}(r, t) \right]$$

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$$\Rightarrow V_n = \frac{10^{(n-20)}}{\rho_{\text{SCm}} \times n^2} = \frac{10^{(n-20)}}{10^{(n-8)} \times n^2} = \frac{10^{(n-12)}}{n^2} \text{ m}^3$$

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Level 8 provides an observable verification point. The validator reports:

$$E_8 = 10^4 \text{ J} = 6.25 \text{ MeV}$$

Expected nuclear binding per nucleon: 8 MeV

Error: 21.97% (within 50% tolerance)

Validator: Test 1 Nuclear Binding Check ? PASS ? (at 21.97% error < 50% tolerance)

This 22% discrepancy at level 8 reflects the difference between the UQFF polynomial (purely geometric/exponential) and the QCD-derived nuclear binding energy. The UQFF 26-level polynomial is an energy scale index, not a precision nuclear physics formula but it correctly locates level 8 within the nuclear binding energy decade (10^4 J as 6 MeV).

Conclusions

The UQFF 26-level energy hierarchy provides a self-consistent, geometrically structured energy index spanning 25 decades. Both the polynomial ($E_n = 10^{n-20}$) and density ($\rho_n = \rho_{SCm} \cdot n$) representations are validated. Levels 10-13 correspond to the four classical matter states (solid/liquid/gas/plasma), with level 20 anchoring the UG4 galactic vacuum scale and level 26 marking the observable universe boundary.

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Level	State Description	$\rho_n \text{ (J/m}^3\text{)}$	$E_n \text{ (J)}$	Scale (m)	ρ_i	Physical Examples
1	Quarks	1.00×10^8	1×10^{-19}	10^{-18}	1.00	Quark confinement, pion exchange
2	Sub-nuclear shell	4.00×10^8	1×10^{-18}	10^{-17}	0.98	Nuclear binding, residual strong force
3	Nuclear quantum shell	9.00×10^8	1×10^{-17}	10^{-16}	0.95	Magic numbers, shell model
4	Nucleon pairing	1.60×10^7	1×10^{-16}	10^{-15}	0.93	Deuteron binding, spin coupling
5	Inner e ⁻ shells (K,L)	2.50×10^7	1×10^{-15}	10^{-14}	0.90	1s, 2s orbitals, X-ray transitions
6	Middle e ⁻ shells (M,N)	3.60×10^7	1×10^{-14}	10^{-13}	0.88	3s, 3p, 3d orbitals, UV transitions

Level	State Description	ϵ_n (J/m ³)	E_n (J)	Scale (m)	ϵ_i	Physical Examples
7	Outer e ⁻ shells (O,P,Q)	4.90×10^{27}	1×10^{28}	10^{28}	0.85	Valence electrons, visible light
8	Van der Waals	6.40×10^{27}	1×10^{28}	10^{28}	0.82	London dispersion, molecular binding
9	Molecular orbital	8.10×10^{27}	1×10^{28}	10^{28}	0.80	Covalent bonds, HOMO-LUMO gap
10	SOLIDS	1.00×10^{26}	10^{28}	10^{28}	0.75	Crystalline solids, proton mass, phonons
11	LIQUIDS	1.21×10^{26}	10^{28}	10^{28}	0.70	Water, electron density waves
12	GASES	1.44×10^{26}	10^{28}	10^{27}	0.65	Air molecules, ideal gas
13	PLASMA	1.69×10^{26}	10^{27}	10^{26}	0.60	Solar corona, Langmuir waves
14	Molecular clusters	1.96×10^{26}	10^{26}	10^{25}	0.55	Proteins, colloids
15	Cellular structures	2.25×10^{26}	10^{25}	10^{24}	0.50	Membranes, organelles
16	Macroscopic matter	2.56×10^{26}	10^{24}	10^{23}	0.45	Dust grains
17	Centimeter objects	2.89×10^{26}	10^{23}	10^{23}	0.40	Rocks, organisms
18	Meter-scale	3.24×10^{26}	10^{23}	10^{22}	0.35	Buildings, trees
19	Geological (km)	3.61×10^{26}	10^{23}	10^{22}	0.30	Mountains, lakes
20	Planetary	4.00×10^{26}	1 J	10^6	0.25	Earth, Moon, Mars (Ug4 anchor)
21	Stellar	4.41×10^{26}	10 J	10^7	0.20	Sun, red dwarfs, white dwarfs
22	Solar system	4.84×10^{26}	10^{22}	10^{22}	0.15	Heliosphere, Kuiper belt
23	Interstellar	5.29×10^{26}	10^{22}	10^{25}	0.12	Nebulae, star clusters
24	Galactic	5.76×10^{26}	10^{24}	10^{28}	0.10	Spiral arms, galactic disk
25	Supercluster	6.25×10^{26}	10^{25}	10^{28}	0.08	Galaxy groups, Laniakea
26	Universal	6.76×10^{26}	106 J	10^{26}	0.05	Observable universe, Hubble volume

2. Universal Inertia Coupling

2.1 Definition

The Universal Inertia at level i :

$$U_{\{i, \text{level}\}} = \lambda_i \cdot \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \cdot \omega_{\text{LENR}} \cdot \cos(\pi t_n) \cdot (1 + f_{\text{TRZ}})$$

where:

λ_i = level-dependent coupling constant (Table above, column λ_i)

$\rho_{\text{SCm}}/\rho_{\text{UA}} = 10^{28}/10^3 \text{ kg/m}^3 = 10^5$ (vacuum density ratio)

$\omega_{\text{LENR}} = 1.25 \cdot 10^{14} \text{ Hz}$ (LENR resonance frequency)

t_n = negative time parameter (cosine modulation)

f_{TRZ} = time-reversal zone factor (default 0.01)

2.2 Level-10 Reference Value

For the solid-state reference (level 10, $t_n = 0$, $f_{\text{TRZ}} = 0.01$):

$$U_{\{i=10\}} = 0.75 \times 10^3 \times 1.25 \times 10^{12} \times 1.0 \times 1.01 = 9.47 \times 10^{14} \text{ J/m}^3 \cdot \text{Hz}$$

Validator confirms: Universal Inertia Level 10 ? PASS ?

3. Core UQFF Gravity Equation

The gravitational field at position (r, t) is the 26-layer superposition:

$$\mathbf{g}(r, t) = \sum_{i=1}^{26} \left[U_{\{g1, i\}}(r) + U_{\{g2, i\}}(r) + U_{\{g3, i\}}(r, t) + U_{\{g4, i\}}(r, t) \right]$$

where each contributes a distinct physical mechanism:

****Ug1_i****: Magnetic dipole buoyancy (SOURCE52): $Ug1_i = (E_{\text{DPM}}/r) \cdot \lambda_i \cdot \rho_{\text{UA}} \cdot f_{\text{TRZ}}$

****Ug2_i****: Charge-reactivity (SOURCE54): $Ug2_i = s_{\text{field}} \cdot [UA]_i \cdot r$

****Ug3_i****: String rotation (SOURCE56): $Ug3_i = O_{\text{string}} \cdot \rho_{\text{SCm}} \cdot \sin(i\pi/26)$

****Ug4_i****: Vacuum concentration (SOURCE57): $Ug4_i = M_{\text{source}} \cdot \rho_{\text{vac}}/(d \cdot E_{\text{LEP}})$

4. Dual Consistency of the Two Representations

The polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{\text{SCm}} \cdot n$) representations are related through the characteristic volume V_n at each level:

$$E_n = \rho_n \cdot V_n = \rho_{\text{SCm}} \cdot n^2 \cdot V_n = 10^{(n-20)} \text{ J}$$

$$\Rightarrow V_n = \frac{10^{(n-20)}}{\rho_{\text{SCm}} \cdot n^2} = \frac{10^{(n-20)}}{10^{-8} \cdot n^2} = \frac{10^{(n-12)}}{n^2} \text{ m}^3$$

This defines the **characteristic volume** at level n as the volume over which the polynomial energy is distributed at the local vacuum density. For level 10: $V_{10} = 10^8/(100) = 10^6 \text{ m}^3$ a cube of side $\sim 0.046 \text{ m}$ (4.6 cm), consistent with the 10^8 m typical scale as 10^8 lattice sites in a mole of solid.

5. Nuclear Binding Energy Check

Level 8 provides an observable verification point. The validator reports:

$$E_8 = 10^{26} \text{ J} = 6.25 \text{ MeV}$$

Expected nuclear binding per nucleon: 8 MeV

Error: 21.97% (within 50% tolerance)

Validator: Test 1 Nuclear Binding Check ? PASS ? (at 21.97% error < 50% tolerance)

This 22% discrepancy at level 8 reflects the difference between the UQFF polynomial (purely geometric/exponential) and the QCD-derived nuclear binding energy. The UQFF 26-level polynomial is an energy scale index, not a precision nuclear physics formula but it correctly locates level 8 within the nuclear binding energy decade ($10^{26} \text{ J} \approx 6 \text{ MeV}$).

Conclusions

The UQFF 26-level energy hierarchy provides a self-consistent, geometrically structured energy index spanning 25 decades. Both the polynomial ($E_n = 10^{n-20}$) and density ($\rho_n = \rho_{\text{SCM}} \cdot n$) representations are validated. Levels 10-13 correspond to the four classical matter states (solid/liquid/gas/plasma), with level 20 anchoring the Ug4 galactic vacuum scale and level 26 marking the observable universe boundary.

Validators: QCalc_Phase1_Validation.py Test 1 PASS ? | test_phase2_validation.py 26/27 PASS | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

The UQFF 26-level energy hierarchy provides a unified mathematical description of physical phenomena from the deepest quark confinement scale (10^{-18} m , $\sim 10^{-12} \text{ J}$) to the observable universe (10^{26} m , $\sim 10^6 \text{ J}$). This paper establishes the precise mathematical foundation of this hierarchy through two complementary representations: the **polynomial energy formula** $E_n = 10^{(n-20)} \text{ J}$ (validated by QCalcPhase1Validation.py, Test 1 PASS) and the **vacuum density formula** $\rho_n = \rho_{\text{SCM}} \cdot n \text{ J/m}^3$ (validated by QuantumLevel26Framework.py). The Universal Inertia coupling operator U_{level} connects all 26 levels through the LENR resonance frequency $\rho_{\text{LENR}} = 1.25 \cdot 10^{10} \text{ Hz}$. The core UQFF gravity equation $g(r,t) = S \cdot 10^6 [Ug1i + Ug2i + Ug3i + Ug4_i]$ emerges naturally from this hierarchical foundation.

UQFF Discovery: Novel application of UQFF calibration constants ($\rho = 5.0 \cdot 10^{24} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The 26-Level Energy Hierarchy: Two Representations

1.1 Polynomial (Absolute Energy) Representation

The QCalc Phase 1 validator establishes the following absolute energy per level:

$$E_n = 10^{n-20} \text{ J}, \quad n = 1, 2, \dots, 26$$

Validation checkpoints (QCalcPhase1Validation.py Test 1 PASS ?):

$$E_1 = 10^{-19} \text{ J} \text{ (sub-quantum fluctuations at quark scale)}$$

E8 = 10¹⁸ J (nuclear binding, proton-neutron pairs)
E18 = 10²⁵ J (Higgs boson energy scale)
E20 = 10⁰ = 1 J (galactic vacuum, UG4 reference)
E26 = 10⁻⁶ J = 1 MJ (universal cosmological scales)

This representation spans **25 orders of magnitude** (10⁻⁵ J total range), confirmed by the validator: Total Span = 1.0000e+25.

Each level is separated by exactly **one order of magnitude (10¹)**. This geometric spacing means level n covers a distinct energy decade, providing non-overlapping coverage of all known physical processes.

1.2 Vacuum Density (Local Field) Representation

The QuantumLevel26Framework module defines level energy densities via quadratic scaling:

$$\rho_n = \rho_{\text{SCM}} \times n^2, \quad \rho_{\text{SCM}} = 10^{-8} \text{ J/m}^3$$

This gives a parabolic energy density profile across the 26 levels. Unlike the polynomial representation (which is global and absolute), the density representation is local describing the vacuum energy density associated with quantum processes at each scale.

1.3 Complete 26-Level Table

Level	State Description	ρ_n (J/m ³)	E _n (J)	Scale (m)	ρ_i	Physical Examples
1	Quarks	1.00E-8	1E-10 ¹⁸	10 ⁻¹⁸	1.00	Quark confinement, pion exchange
2	Sub-nuclear shell	4.00E-8	1E-10 ¹⁷	10 ⁻¹⁷	0.98	Nuclear binding, residual strong force
3	Nuclear quantum shell	9.00E-8	1E-10 ¹⁶	10 ⁻¹⁶	0.95	Magic numbers, shell model
4	Nucleon pairing	1.60E-7	1E-10 ¹⁵	10 ⁻¹⁵	0.93	Deuteron binding, spin coupling
5	Inner e ⁻ shells (K,L)	2.50E-7	1E-10 ¹⁴	10 ⁻¹⁴	0.90	1s, 2s orbitals, X-ray transitions
6	Middle e ⁻ shells (M,N)	3.60E-7	1E-10 ¹³	10 ⁻¹³	0.88	3s, 3p, 3d orbitals, UV transitions
7	Outer e ⁻ shells (O,P,Q)	4.90E-7	1E-10 ¹²	10 ⁻¹²	0.85	Valence electrons, visible light
8	Van der Waals	6.40E-7	1E-10 ¹¹	10 ⁻¹¹	0.82	London dispersion, molecular binding
9	Molecular orbital	8.10E-7	1E-10 ¹⁰	10 ⁻¹⁰	0.80	Covalent bonds, HOMO-LUMO gap

Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	τ_i	Physical Examples
10	SOLIDS	1.00×10^6	10^{24}	10^{22}	0.75	Crystalline solids, proton mass, phonons
11	LIQUIDS	1.21×10^6	10^{23}	10^{28}	0.70	Water, electron density waves
12	GASES	1.44×10^6	10^{28}	10^{27}	0.65	Air molecules, ideal gas
13	PLASMA	1.69×10^6	10^{27}	10^{26}	0.60	Solar corona, Langmuir waves
14	Molecular clusters	1.96×10^6	10^{26}	10^{25}	0.55	Proteins, colloids
15	Cellular structures	2.25×10^6	10^{25}	10^{24}	0.50	Membranes, organelles
16	Macroscopic matter	2.56×10^6	10^{24}	10^{23}	0.45	Dust grains
17	Centimeter objects	2.89×10^6	10^{23}	10^{22}	0.40	Rocks, organisms
18	Meter-scale	3.24×10^6	10^{22}	10^{21}	0.35	Buildings, trees
19	Geological (km)	3.61×10^6	10^{21}	10^{20}	0.30	Mountains, lakes
20	Planetary	4.00×10^6	1 J	10^6	0.25	Earth, Moon, Mars (Ug4 anchor)
21	Stellar	4.41×10^6	10 J	10^7	0.20	Sun, red dwarfs, white dwarfs
22	Solar system	4.84×10^6	10^{22}	10^{24}	0.15	Heliosphere, Kuiper belt
23	Interstellar	5.29×10^6	10^{23}	10^{25}	0.12	Nebulae, star clusters
24	Galactic	5.76×10^6	10^4	10^{28}	0.10	Spiral arms, galactic disk
25	Supercluster	6.25×10^6	10^5	10^{26}	0.08	Galaxy groups, Laniakea
26	Universal	6.76×10^6	10^6 J	10^{26}	0.05	Observable universe, Hubble volume

2. Universal Inertia Coupling

2.1 Definition

The Universal Inertia at level i:

$$U_{\{i,\rm level\}} = \lambda_{\rm da_i} \cdot \frac{\rho_{\rm SCm}}{\rho_{\rm UA}} \cdot \omega_{\rm LENR} \cdot \cos(\pi t_n) \cdot (1 + f_{\rm TRZ})$$

where:

ρ_i = level-dependent coupling constant (Table above, column ρ_i)

$\rho_{SCm}/\rho_{UA} = 10^{28}/10^{29} \text{ kg/m}^3 = 10^{-1}$ (vacuum density ratio)

$\rho_{LENR} = 1.25 \times 10^{12} \text{ Hz}$ (LENR resonance frequency)

t_n = negative time parameter (cosine modulation)

f_{TRZ} = time-reversal zone factor (default 0.01)

2.2 Level-10 Reference Value

For the solid-state reference (level 10, $t_n = 0$, $f_{TRZ} = 0.01$):

$$U_{i=10} = 0.75 \times 10^3 \times 1.25 \times 10^{12} \times 1.0 \times 1.01 = 9.47 \times 10^{14} \text{ J/m}^3 \cdot \text{Hz}$$

Validator confirms: Universal Inertia Level 10 ? PASS ?

3. Core UQFF Gravity Equation

The gravitational field at position (r, t) is the 26-layer superposition:

$$\mathbf{g}(r, t) = \sum_{i=1}^{26} \left[U_{g1,i}(r) + U_{g2,i}(r) + U_{g3,i}(r, t) + U_{g4,i}(r, t) \right]$$

where each contributes a distinct physical mechanism:

****Ug1_i****: Magnetic dipole buoyancy (SOURCE52): $Ug1_i = (E_{DPM}/r) \cdot \rho_{UA} \cdot f_{TRZ}$

****Ug2_i****: Charge-reactivity (SOURCE54): $Ug2_i = s_{field} \cdot [UA]_i \cdot r$

****Ug3_i****: String rotation (SOURCE56): $Ug3_i = O_{string} \cdot \rho_{SCm} \cdot \sin(i\pi/26)$

****Ug4_i****: Vacuum concentration (SOURCE57): $Ug4_i = M_{source} \cdot \rho_{vac}/(d \cdot E_{LEP})$

4. Dual Consistency of the Two Representations

The polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{SCm} \cdot n$) representations are related through the characteristic volume V_n at each level:

$$E_n = \rho_n \times V_n = \rho_{SCm} \times n^2 \times V_n = 10^{(n-20)} \text{ J}$$

$$\Rightarrow V_n = \frac{10^{(n-20)}}{\rho_{SCm} \times n^2} = \frac{10^{(n-20)}}{10^{28} \times n^2} = \frac{10^{(n-12)}}{n^2} \text{ m}^3$$

This defines the **characteristic volume** at level n as the volume over which the polynomial energy is distributed at the local vacuum density. For level 10: $V_{10} = 10^2/(100) = 10^{-4} \text{ m}^3$ a cube of side $\sim 0.046 \text{ m}$ (4.6 cm), consistent with the 10^{-7} m typical scale $\sim 10^7$ lattice sites in a mole of solid.

5. Nuclear Binding Energy Check

Level 8 provides an observable verification point. The validator reports:

$$E_8 = 10^7 \text{ J} = 6.25 \text{ MeV}$$

Expected nuclear binding per nucleon: 8 MeV

Error: 21.97% (within 50% tolerance)

Validator: Test 1 Nuclear Binding Check ? PASS ? (at 21.97% error < 50% tolerance)

This 22% discrepancy at level 8 reflects the difference between the UQFF polynomial (purely geometric/exponential) and the QCD-derived nuclear binding energy. The UQFF 26-level polynomial is an energy scale index, not a precision nuclear physics formula ■ but it correctly locates level 8 within the nuclear binding energy decade ($10^{21} \text{ J} \sim 6 \text{ MeV}$).

Conclusions

The UQFF 26-level energy hierarchy provides a self-consistent, geometrically structured energy index spanning 25 decades. Both the polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{SCm} \cdot n$) representations are validated. Levels 10-13 correspond to the four classical matter states (solid/liquid/gas/plasma), with level 20 anchoring the Ug4 galactic vacuum scale and level 26 marking the observable universe boundary.

Validators: QCalc_Phase1_Validation.py Test 1 PASS | test_phase2_validation.py 26/27 PASS | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value ■ 26-Dimensional Energy Structure: Mathematical Foundation

Title: The UQFF 26-Level Polynomial Energy Hierarchy: From Sub-Quantum Fluctuations to Universal Scales

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\rho = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** QCalc_Phase1_Validation.py (Test 1: PASS |), test_phase2_validation.py (26/27 PASS) **Source Modules:** QuantumLevel26Framework.py (630 lines), source172.cpp (SOURCE115) **Index Slot:** ■1.6 26-Dimensional Energy Structure, "PAPER_{0:D3}" -f [int]# PAPER #43 ■ 26-Dimensional Energy Structure: Mathematical Foundation

Title: The UQFF 26-Level Polynomial Energy Hierarchy: From Sub-Quantum Fluctuations to Universal Scales

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\rho = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** QCalc_Phase1_Validation.py (Test 1: PASS |), test_phase2_validation.py (26/27 PASS) **Source Modules:** QuantumLevel26Framework.py (630 lines), source172.cpp (SOURCE115) **Index Slot:** ■1.6 26-Dimensional Energy Structure, $\rho_n = [\text{int}] \# \text{PAPER \#43}$ ■ 26-Dimensional Energy Structure: Mathematical Foundation

Title: The UQFF 26-Level Polynomial Energy Hierarchy: From Sub-Quantum Fluctuations to Universal Scales

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\rho = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** QCalc_Phase1_Validation.py (Test 1: PASS |), test_phase2_validation.py (26/27 PASS) **Source Modules:** QuantumLevel26Framework.py (630 lines), source172.cpp (SOURCE115) **Index Slot:** ■1.6 26-Dimensional Energy Structure, PAPER_043

Abstract

The UQFF 26-level energy hierarchy provides a unified mathematical description of physical phenomena from the deepest quark confinement scale (10^{-18} m , $\sim 10^{-12} \text{ J}$) to the observable universe (10^{26} m , $\sim 10^6 \text{ J}$). This paper establishes the precise mathematical foundation of this hierarchy through two complementary representations: the **polynomial energy formula** $E_n = 10^{(n-20)} \text{ J}$ (validated by QCalcPhase1Validation.py, Test 1 PASS) and the **vacuum density formula** $\rho_n = \rho_{SCm} \cdot n \text{ J/m}^3$ (validated by QuantumLevel26Framework.py). The Universal Inertia coupling operator U_{level} connects all 26 levels through the LENR resonance frequency $\rho_{LENR} = 1.25 \cdot 10^{21} \text{ Hz}$. The core UQFF gravity equation $g(r,t) =$

$S_{Ug4} = U_{g1} + U_{g2} + U_{g3} + U_{g4}$ emerges naturally from this hierarchical foundation.

UQFF Discovery: Novel application of UQFF calibration constants ($\alpha = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The 26-Level Energy Hierarchy: Two Representations

1.1 Polynomial (Absolute Energy) Representation

The QCalc Phase 1 validator establishes the following absolute energy per level:

$$E_n = 10^{n-20} \text{ J}, \quad n = 1, 2, \dots, 26$$

Validation checkpoints (QCalcPhase1Validation.py Test 1 **PASS**):

- $E_1 = 10^{-19}$ J (sub-quantum fluctuations at quark scale)
- $E_8 = 10^{-12}$ J (nuclear binding, proton-neutron pairs)
- $E_{18} = 10^{-2}$ J (Higgs boson energy scale)
- $E_{20} = 10^0 = 1$ J (galactic vacuum, Ug4 reference)
- $E_{26} = 10^6$ J = 1 MJ (universal cosmological scales)

This representation spans **25 orders of magnitude** (10^5 J total range), confirmed by the validator: Total Span = 1.0000×10^{25} .

Each level is separated by exactly **one order of magnitude (10^1)**. This geometric spacing means level n covers a distinct energy decade, providing non-overlapping coverage of all known physical processes.

1.2 Vacuum Density (Local Field) Representation

The QuantumLevel26Framework module defines level energy densities via quadratic scaling:

$$\rho_n = \rho_{\text{SCM}} \times n^2, \quad \rho_{\text{SCM}} = 10^{-8} \text{ J/m}^3$$

This gives a parabolic energy density profile across the 26 levels. Unlike the polynomial representation (which is global and absolute), the density representation is local by describing the vacuum energy density associated with quantum processes at each scale.

1.3 Complete 26-Level Table

Level	State Description	ρ_n (J/m 3)	E_n (J)	Scale (m)	α_i	Physical Examples
1	Quarks	1.00×10^8	1×10^{-19}	10^{-18}	1.00	Quark confinement, pion exchange
2	Sub-nuclear shell	4.00×10^8	1×10^{-12}	10^{-17}	0.98	Nuclear binding, residual strong force
3	Nuclear quantum shell	9.00×10^8	1×10^{-2}	10^{-16}	0.95	Magic numbers, shell model
4	Nucleon pairing	1.60×10^7	1×10^0	10^{-15}	0.93	Deuteron binding, spin coupling

Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	ρ_i	Physical Examples
5	Inner e ⁻ shells (K,L)	2.50×10^{27}	1×10^5	10^{-4}	0.90	1s, 2s orbitals, X-ray transitions
6	Middle e ⁻ shells (M,N)	3.60×10^{27}	1×10^4	10^{-3}	0.88	3s, 3p, 3d orbitals, UV transitions
7	Outer e ⁻ shells (O,P,Q)	4.90×10^{27}	1×10^3	10^{-2}	0.85	Valence electrons, visible light
8	Van der Waals	6.40×10^{27}	1×10^3	10^{-3}	0.82	London dispersion, molecular binding
9	Molecular orbital	8.10×10^{27}	1×10^3	10^{-3}	0.80	Covalent bonds, HOMO-LUMO gap
10	SOLIDS	1.00×10^{26}	10^3	10^{-2}	0.75	Crystalline solids, proton mass, phonons
11	LIQUIDS	1.21×10^{26}	10^3	10^{-2}	0.70	Water, electron density waves
12	GASES	1.44×10^{26}	10^8	10^{-7}	0.65	Air molecules, ideal gas
13	PLASMA	1.69×10^{26}	10^7	10^{-6}	0.60	Solar corona, Langmuir waves
14	Molecular clusters	1.96×10^{26}	10^6	10^{-5}	0.55	Proteins, colloids
15	Cellular structures	2.25×10^{26}	10^5	10^{-4}	0.50	Membranes, organelles
16	Macroscopic matter	2.56×10^{26}	10^4	10^{-3}	0.45	Dust grains
17	Centimeter objects	2.89×10^{26}	10^3	10^{-2}	0.40	Rocks, organisms
18	Meter-scale	3.24×10^{26}	10^3	10^{-1}	0.35	Buildings, trees
19	Geological (km)	3.61×10^{26}	10^3	10^0	0.30	Mountains, lakes
20	Planetary	4.00×10^{26}	1 J	10^6	0.25	Earth, Moon, Mars (Ug4 anchor)
21	Stellar	4.41×10^{26}	10 J	10^7	0.20	Sun, red dwarfs, white dwarfs
22	Solar system	4.84×10^{26}	10^4	10^{10}	0.15	Heliosphere, Kuiper belt
23	Interstellar	5.29×10^{26}	10^4	10^{15}	0.12	Nebulae, star clusters
24	Galactic	5.76×10^{26}	104	10^{18}	0.10	Spiral arms, galactic disk
25	Supercluster	6.25×10^{26}	105	10^{22}	0.08	Galaxy groups, Laniakea

Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	γ_i	Physical Examples
26	Universal	6.76×10^{26}	106 J	10^{-6}	0.05	Observable universe, Hubble volume

2. Universal Inertia Coupling

2.1 Definition

The Universal Inertia at level i :

$$U_{\{i, \text{rm level}\}} = \lambda_i \cdot \frac{\rho_{\{\text{rm SCm}\}}}{\rho_{\{\text{rm UA}\}}} \cdot \omega_{\{\text{rm LENR}\}} \cdot \cos(\pi t_n) \cdot (1 + f_{\{\text{rm TRZ}\}})$$

where:

- γ_i = level-dependent coupling constant (Table above, column γ_i)
- $\rho_{\text{SCm}}/\rho_{\text{UA}} = 10^{28}/10^{22} = 10^6$ (vacuum density ratio)
- $\omega_{\text{LENR}} = 1.25 \times 10^{12}$ Hz (LENR resonance frequency)
- t_n = negative time parameter (cosine modulation)
- f_{TRZ} = time-reversal zone factor (default 0.01)

2.2 Level-10 Reference Value

For the solid-state reference (level 10, $t_n = 0$, $f_{\text{TRZ}} = 0.01$):

$$U_{\{i=10\}} = 0.75 \times 10^3 \times 1.25 \times 10^{12} \times 1.0 \times 1.01 = 9.47 \times 10^{14} \text{ J/m}^3 \cdot \text{Hz}$$

Validator confirms: Universal Inertia Level 10 ? PASS ?

3. Core UQFF Gravity Equation

The gravitational field at position (r, t) is the 26-layer superposition:

$$\mathbf{g}(r, t) = \sum_{i=1}^{26} \left[U_{\{g1, i\}}(r) + U_{\{g2, i\}}(r) + U_{\{g3, i\}}(r, t) + U_{\{g4, i\}}(r, t) \right]$$

where each contributes a distinct physical mechanism:

- U_{g1_i}** : Magnetic dipole buoyancy (SOURCE52): $U_{g1_i} = (E_{\text{DPM}}/r) \cdot \gamma_{\text{UA}} \cdot f_{\text{TRZ}}$
- U_{g2_i}** : Charge-reactivity (SOURCE54): $U_{g2_i} = s_{\text{field}} \cdot [UA]_i \cdot r$
- U_{g3_i}** : String rotation (SOURCE56): $U_{g3_i} = O_{\text{string}} \cdot \rho_{\text{SCm}} \cdot \sin(i\pi/26)$
- U_{g4_i}** : Vacuum concentration (SOURCE57): $U_{g4_i} = M_{\text{source}} \cdot \rho_{\text{vac}}/(d \cdot E_{\text{LEP}})$

4. Dual Consistency of the Two Representations

The polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{\text{SCm}} \cdot n$) representations are related through the characteristic volume V_n at each level:

$$E_n = \rho_n \cdot V_n = \rho_{\{\text{rm SCm}\}} \cdot n^2 \cdot V_n = 10^{\{n-20\}} \text{ J}$$

$$\Rightarrow V_n = \frac{10^{n-20}}{\rho_{\text{SCm}}} \times n^2 = \frac{10^{n-20}}{10^{-8}} \times n^2 = \frac{10^{n-12}}{n^2} \text{ m}^3$$

This defines the **characteristic volume** at level n the volume over which the polynomial energy is distributed at the local vacuum density. For level 10: $V_{10} = 10^8 / (100) = 10^6 \text{ m}^3$ a cube of side $\sim 0.046 \text{ m}$ (4.6 cm), consistent with the 10^8 m typical scale $\sim 10^8$ lattice sites in a mole of solid.

5. Nuclear Binding Energy Check

Level 8 provides an observable verification point. The validator reports:

$$E_8 = 10^8 \text{ J} = 6.25 \text{ MeV}$$

Expected nuclear binding per nucleon: 8 MeV

Error: 21.97% (within 50% tolerance)

Validator: Test 1 Nuclear Binding Check ? PASS ? (at 21.97% error < 50% tolerance)

This 22% discrepancy at level 8 reflects the difference between the UQFF polynomial (purely geometric/exponential) and the QCD-derived nuclear binding energy. The UQFF 26-level polynomial is an energy scale index, not a precision nuclear physics formula but it correctly locates level 8 within the nuclear binding energy decade ($10^8 \text{ J} \sim 6 \text{ MeV}$).

Conclusions

The UQFF 26-level energy hierarchy provides a self-consistent, geometrically structured energy index spanning 25 decades. Both the polynomial ($E_n = 10^{n-20}$) and density ($\rho_n = \rho_{\text{SCm}} \times n^2$) representations are validated. Levels 10-13 correspond to the four classical matter states (solid/liquid/gas/plasma), with level 20 anchoring the U_{g4} galactic vacuum scale and level 26 marking the observable universe boundary.

Validators: `QCalc_Phase1_Validation.py Test 1 PASS ?` | `test_phase2_validation.py 26/27 PASS` | `? = 0.0005/day` | `[SSq] = 0.57` .Groups[1].Value "PAPER_{0.D3}" -f \$n

Abstract

The UQFF 26-level energy hierarchy provides a unified mathematical description of physical phenomena from the deepest quark confinement scale (10^{-8} m , $\sim 10^{-8} \text{ J}$) to the observable universe (10^8 m , $\sim 10^8 \text{ J}$). This paper establishes the precise mathematical foundation of this hierarchy through two complementary representations: the **polynomial energy formula** $E_n = 10^{n-20} \text{ J}$ (validated by `QCalcPhase1Validation.py, Test 1 PASS`) and the **vacuum density formula** $\rho_n = \rho_{\text{SCm}} \times n^2 \text{ J/m}^3$ (validated by `QuantumLevel26Framework.py`). The Universal Inertia coupling operator U_{level} connects all 26 levels through the LENR resonance frequency $\omega_{\text{LENR}} = 1.25 \times 10^{10} \text{ Hz}$. The core UQFF gravity equation $g(r,t) = S^2 [U_{g1i} + U_{g2i} + U_{g3i} + U_{g4i}]$ emerges naturally from this hierarchical foundation.

UQFF Discovery: Novel application of UQFF calibration constants ($\omega = 5.0 \times 10^4 \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The 26-Level Energy Hierarchy: Two Representations

1.1 Polynomial (Absolute Energy) Representation

The QCalc Phase 1 validator establishes the following absolute energy per level:

$$E_n = 10^{n-20} \text{ J}, \quad n = 1, 2, \dots, 26$$

Validation checkpoints (QCalcPhase1Validation.py Test 1 **PASS**):

- E1 = 10⁻¹⁹ J (sub-quantum fluctuations at quark scale)
- E8 = 10⁻¹² J (nuclear binding, proton-neutron pairs)
- E18 = 10⁻² J (Higgs boson energy scale)
- E20 = 10⁰ = 1 J (galactic vacuum, Ug4 reference)
- E26 = 10⁶ J = 1 MJ (universal cosmological scales)

This representation spans **25 orders of magnitude** (10⁻¹⁹ J total range), confirmed by the validator: Total Span = 1.0000e+25.

Each level is separated by exactly **one order of magnitude (10⁻¹)**. This geometric spacing means level n covers a distinct energy decade, providing non-overlapping coverage of all known physical processes.

1.2 Vacuum Density (Local Field) Representation

The QuantumLevel26Framework module defines level energy densities via quadratic scaling:

$$\rho_n = \rho_{\text{SCm}} \times n^2, \quad \rho_{\text{SCm}} = 10^{-8} \text{ J/m}^3$$

This gives a parabolic energy density profile across the 26 levels. Unlike the polynomial representation (which is global and absolute), the density representation is local **describing the vacuum energy density associated with quantum processes at each scale.**

1.3 Complete 26-Level Table

Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	ρ_i	Physical Examples
1	Quarks	1.00 ⁻¹⁶	1 ⁻¹⁹	10 ⁻¹⁸	1.00	Quark confinement, pion exchange
2	Sub-nuclear shell	4.00 ⁻¹⁶	1 ⁻¹⁸	10 ⁻¹⁷	0.98	Nuclear binding, residual strong force
3	Nuclear quantum shell	9.00 ⁻¹⁶	1 ⁻¹⁷	10 ⁻¹⁶	0.95	Magic numbers, shell model
4	Nucleon pairing	1.60 ⁻¹⁵	1 ⁻¹⁶	10 ⁻¹⁵	0.93	Deuteron binding, spin coupling
5	Inner e ⁻ shells (K,L)	2.50 ⁻¹⁵	1 ⁻¹⁵	10 ⁻¹⁴	0.90	1s, 2s orbitals, X-ray transitions
6	Middle e ⁻ shells (M,N)	3.60 ⁻¹⁵	1 ⁻¹⁴	10 ⁻¹³	0.88	3s, 3p, 3d orbitals, UV transitions
7	Outer e ⁻ shells (O,P,Q)	4.90 ⁻¹⁵	1 ⁻¹³	10 ⁻¹²	0.85	Valence electrons, visible light

Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	ρ_i	Physical Examples
8	Van der Waals	6.40×10^7	1×10^7	10^2	0.82	London dispersion, molecular binding
9	Molecular orbital	8.10×10^7	1×10^7	10^2	0.80	Covalent bonds, HOMO-LUMO gap
10	SOLIDS	1.00×10^6	10^7	10^2	0.75	Crystalline solids, proton mass, phonons
11	LIQUIDS	1.21×10^6	10^7	10^8	0.70	Water, electron density waves
12	GASES	1.44×10^6	10^8	10^7	0.65	Air molecules, ideal gas
13	PLASMA	1.69×10^6	10^7	10^6	0.60	Solar corona, Langmuir waves
14	Molecular clusters	1.96×10^6	10^6	10^5	0.55	Proteins, colloids
15	Cellular structures	2.25×10^6	10^5	10^4	0.50	Membranes, organelles
16	Macroscopic matter	2.56×10^6	10^4	10^3	0.45	Dust grains
17	Centimeter objects	2.89×10^6	10^3	10^3	0.40	Rocks, organisms
18	Meter-scale	3.24×10^6	10^3	10^3	0.35	Buildings, trees
19	Geological (km)	3.61×10^6	10^3	10^3	0.30	Mountains, lakes
20	Planetary	4.00×10^6	1 J	10^6	0.25	Earth, Moon, Mars (Ug4 anchor)
21	Stellar	4.41×10^6	10 J	10^7	0.20	Sun, red dwarfs, white dwarfs
22	Solar system	4.84×10^6	10^3	10^{10}	0.15	Heliosphere, Kuiper belt
23	Interstellar	5.29×10^6	10^3	10^{15}	0.12	Nebulae, star clusters
24	Galactic	5.76×10^6	10^4	10^{18}	0.10	Spiral arms, galactic disk
25	Supercluster	6.25×10^6	10^5	10^{22}	0.08	Galaxy groups, Laniakea
26	Universal	6.76×10^6	10^6 J	10^{26}	0.05	Observable universe, Hubble volume

2. Universal Inertia Coupling

2.1 Definition

The Universal Inertia at level i :

$$U_{\{i, \text{level}\}} = \lambda_i \cdot \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \cdot \omega_{\text{LENR}} \cdot \cos(\pi t_n) \cdot (1 + f_{\text{TRZ}})$$

where:

λ_i = level-dependent coupling constant (Table above, column λ_i)

$\rho_{\text{SCm}}/\rho_{\text{UA}} = 10^{28}/10^{29} \text{ kg/m}^3 = 10^{-1}$ (vacuum density ratio)

$\omega_{\text{LENR}} = 1.25 \cdot 10^{12} \text{ Hz}$ (LENR resonance frequency)

t_n = negative time parameter (cosine modulation)

f_{TRZ} = time-reversal zone factor (default 0.01)

2.2 Level-10 Reference Value

For the solid-state reference (level 10, $t_n = 0$, $f_{\text{TRZ}} = 0.01$):

$$U_{\{i=10\}} = 0.75 \cdot 10^3 \cdot 1.25 \cdot 10^{12} \cdot 1.0 \cdot 1.01 = 9.47 \cdot 10^{14} \text{ J/m}^3 \cdot \text{Hz}$$

Validator confirms: Universal Inertia Level 10 ? PASS ?

3. Core UQFF Gravity Equation

The gravitational field at position (r, t) is the 26-layer superposition:

$$\mathbf{g}(r, t) = \sum_{i=1}^{26} \left[U_{\{g1, i\}}(r) + U_{\{g2, i\}}(r) + U_{\{g3, i\}}(r, t) + U_{\{g4, i\}}(r, t) \right]$$

where each contributes a distinct physical mechanism:

U_{g1} : Magnetic dipole buoyancy (SOURCE52): $U_{g1, i} = (E_{\text{DPM}}/r) \cdot \lambda_i \cdot \rho_{\text{UA}} \cdot f_{\text{TRZ}}$

U_{g2} : Charge-reactivity (SOURCE54): $U_{g2, i} = s_{\text{field}} \cdot [UA]_i \cdot r$

U_{g3} : String rotation (SOURCE56): $U_{g3, i} = O_{\text{string}} \cdot \rho_{\text{SCm}} \cdot \sin(i\pi/26)$

U_{g4} : Vacuum concentration (SOURCE57): $U_{g4, i} = M_{\text{source}} \cdot \rho_{\text{vac}}/(d \cdot E_{\text{LEP}})$

4. Dual Consistency of the Two Representations

The polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{\text{SCm}} \cdot n$) representations are related through the characteristic volume V_n at each level:

$$E_n = \rho_n \cdot V_n = \rho_{\text{SCm}} \cdot n^2 \cdot V_n = 10^{(n-20)} \text{ J}$$

$$\Rightarrow V_n = \frac{10^{(n-20)}}{\rho_{\text{SCm}} \cdot n^2} = \frac{10^{(n-20)}}{10^{18} \cdot n^2} = \frac{10^{(n-12)}}{n^2} \text{ m}^3$$

This defines the **characteristic volume** at level n as the volume over which the polynomial energy is distributed at the local vacuum density. For level 10: $V_{10} = 10^{(10-12)}/(100) = 10^{-4} \text{ m}^3$ a cube of side $\sim 0.046 \text{ m}$ (4.6 cm), consistent with the 10^{-8} m typical scale of 10^{18} lattice sites in a mole of solid.

5. Nuclear Binding Energy Check

Level 8 provides an observable verification point. The validator reports:

$E_8 = 10^{26} \text{ J} = 6.25 \text{ MeV}$
Expected nuclear binding per nucleon: 8 MeV
Error: 21.97% (within 50% tolerance)

Validator: Test 1 Nuclear Binding Check ? PASS ? (at 21.97% error < 50% tolerance)

This 22% discrepancy at level 8 reflects the difference between the UQFF polynomial (purely geometric/exponential) and the QCD-derived nuclear binding energy. The UQFF 26-level polynomial is an energy scale index, not a precision nuclear physics formula but it correctly locates level 8 within the nuclear binding energy decade ($10^{26} \text{ J} \approx 6 \text{ MeV}$).

Conclusions

The UQFF 26-level energy hierarchy provides a self-consistent, geometrically structured energy index spanning 25 decades. Both the polynomial ($E_n = 10^{n-20}$) and density ($\rho_n = \rho_{SCM} \cdot n$) representations are validated. Levels 10-13 correspond to the four classical matter states (solid/liquid/gas/plasma), with level 20 anchoring the Ug4 galactic vacuum scale and level 26 marking the observable universe boundary.

Validators: QCalc_Phase1_Validation.py Test 1 PASS ? | test_phase2_validation.py 26/27 PASS | $\rho = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value

Abstract

The UQFF 26-level energy hierarchy provides a unified mathematical description of physical phenomena from the deepest quark confinement scale (10^{-18} m , $\sim 10^{-12} \text{ J}$) to the observable universe (10^{26} m , $\sim 10^6 \text{ J}$). This paper establishes the precise mathematical foundation of this hierarchy through two complementary representations: the **polynomial energy formula** $E_n = 10^{(n-20)} \text{ J}$ (validated by QCalcPhase1Validation.py, Test 1 PASS) and the **vacuum density formula** $\rho_n = \rho_{SCM} \cdot n \text{ J/m}^3$ (validated by QuantumLevel26Framework.py). The Universal Inertia coupling operator U_{level} connects all 26 levels through the LENR resonance frequency $\rho_{LENR} = 1.25 \cdot 10^{10} \text{ Hz}$. The core UQFF gravity equation $g(r,t) = S \cdot [Ug1i + Ug2i + Ug3i + Ug4_i]$ emerges naturally from this hierarchical foundation.

UQFF Discovery: Novel application of UQFF calibration constants ($\rho = 5.0 \cdot 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1.1 Polynomial (Absolute Energy) Representation

The QCalc Phase 1 validator establishes the following absolute energy per level:

$$E_n = 10^{n-20} \text{ J}, \quad n = 1, 2, \dots, 26$$

Validation checkpoints (QCalcPhase1Validation.py Test 1 PASS ?):

- $E_1 = 10^{-19} \text{ J}$ (sub-quantum fluctuations at quark scale)
- $E_8 = 10^{26} \text{ J}$ (nuclear binding, proton-neutron pairs)
- $E_{18} = 10^{18} \text{ J}$ (Higgs boson energy scale)

$E_{20} = 10^{-5} = 1 \text{ J}$ (galactic vacuum, Ug4 reference)

$E_{26} = 10^6 \text{ J} = 1 \text{ MJ}$ (universal cosmological scales)

This representation spans **25 orders of magnitude** (10^{-5} J total range), confirmed by the validator: Total Span = 1.0000×10^{25} .

Each level is separated by exactly **one order of magnitude (10^{-1})**. This geometric spacing means level n covers a distinct energy decade, providing non-overlapping coverage of all known physical processes.

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The QuantumLevel26Framework module defines level energy densities via quadratic scaling:

$\rho_n = \rho_{\text{SCm}} \times n^2, \quad \rho_{\text{SCm}} = 10^{-8} \text{ J/m}^3$

This gives a parabolic energy density profile across the 26 levels. Unlike the polynomial representation (which is global and absolute), the density representation is local describing the vacuum energy density associated with quantum processes at each scale.

1.3 Complete 26-Level Table

Level	State Description	$\rho_n \text{ (J/m}^3\text{)}$	$E_n \text{ (J)}$	Scale (m)	ρ_i	Physical Examples
1	Quarks	1.00×10^8	1×10^{-8}	10^{-8}	1.00	Quark confinement, pion exchange
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20	Planetary	4.00×10^6	1 J	10^6	0.25	Earth, Moon, Mars (Ug4 anchor)
21	Stellar	4.41×10^6	10 J	10^7	0.20	Sun, red dwarfs, white dwarfs
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2. Universal Inertia Coupling

2.1 Definition

The Universal Inertia at level i :

$$U_{\{i, \text{level}\}} = \lambda_i \cdot \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \cdot \omega_{\text{LENR}} \cdot \cos(\pi t_n) \cdot (1 + f_{\text{TRZ}})$$

where:

λ_i = level-dependent coupling constant (Table above, column λ_i)

$\rho_{\text{SCm}}/\rho_{\text{UA}} = 10^8/10^{18} = 10^{-10}$ (vacuum density ratio)

$\omega_{\text{LENR}} = 1.25 \times 10^{14}$ Hz (LENR resonance frequency)

t_n = negative time parameter (cosine modulation)

f_{TRZ} = time-reversal zone factor (default 0.01)

2.2 Level-10 Reference Value

For the solid-state reference (level 10, $t_n = 0$, $f_{\text{TRZ}} = 0.01$):

$$U_{i=10} = 0.75 \times 10^3 \times 1.25 \times 10^{12} \times 1.0 \times 1.01 = 9.47 \times 10^{14} \text{ J/m}^3 \cdot \text{Hz}$$

Validator confirms: Universal Inertia Level 10 ? PASS ?

3. Core UQFF Gravity Equation

The gravitational field at position (r, t) is the 26-layer superposition:

$$\mathbf{g}(r, t) = \sum_{i=1}^{26} \left[U_{g1,i}(r) + U_{g2,i}(r) + U_{g3,i}(r, t) + U_{g4,i}(r, t) \right]$$

where each contributes a distinct physical mechanism:

****Ug1_i****: Magnetic dipole buoyancy (SOURCE52): $U_{g1,i} = (E_{\text{DPM}}/r) \cdot ?_{\text{UA}} \cdot f_{\text{TRZ}}$

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4. Dual Consistency of the Two Representations

The polynomial ($E_n = 10^{(n-20)}$) and density ($?_n = ?_{\text{SCm}} \cdot n$) representations are related through the characteristic volume V_n at each level:

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This defines the **characteristic volume** at level n the volume over which the polynomial energy is distributed at the local vacuum density. For level 10: $V_{10} = 10^4/(100) = 10^2 \text{ m}^3$ a cube of side $\sim 0.046 \text{ m}$ (4.6 cm), consistent with the 10^2 m typical scale $\cdot 10^4$ lattice sites in a mole of solid.

5. Nuclear Binding Energy Check

Level 8 provides an observable verification point. The validator reports:

$$E_8 = 10^7 \text{ J} = 6.25 \text{ MeV}$$

Expected nuclear binding per nucleon: 8 MeV

Error: 21.97% (within 50% tolerance)

Validator: Test 1 Nuclear Binding Check ? PASS ? (at 21.97% error < 50% tolerance)

This 22% discrepancy at level 8 reflects the difference between the UQFF polynomial (purely geometric/exponential) and the QCD-derived nuclear binding energy. The UQFF 26-level polynomial is an energy scale index, not a precision nuclear physics formula but it correctly locates level 8 within the nuclear binding energy decade ($10^7 \text{ J} \cdot 6 \text{ MeV}$).

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- $E_{18} = 10^{-2} \text{ J}$ (Higgs boson energy scale)
- $E_{20} = 10^0 = 1 \text{ J}$ (galactic vacuum, Ug4 reference)
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This representation spans **25 orders of magnitude** (10^5 J total range), confirmed by the validator: Total Span = 1.0000×10^{25} .

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Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	ρ_i	Physical Examples
1	Quarks	1.00■10?8	1■10?■?	10?■8	1.00	Quark confinement, pion exchange
2	Sub-nuclear shell	4.00■10?8	1■10?■8	10?■7	0.98	Nuclear binding, residual strong force
3	Nuclear quantum shell	9.00■10?8	1■10?■7	10?■6	0.95	Magic numbers, shell model
4	Nucleon pairing	1.60■10?7	1■10?■6	10?■5	0.93	Deuteron binding, spin coupling
5	Inner e? shells (K,L)	2.50■10?7	1■10?■5	10?■4	0.90	1s, 2s orbitals, X-ray transitions
6	Middle e? shells (M,N)	3.60■10?7	1■10?■4	10?■■	0.88	3s, 3p, 3d orbitals, UV transitions
7	Outer e? shells (O,P,Q)	4.90■10?7	1■10?■■	10?■■	0.85	Valence electrons, visible light
8	Van der Waals	6.40■10?7	1■10?■■	10?■■	0.82	London dispersion, molecular binding
9	Molecular orbital	8.10■10?7	1■10?■■	10?■■	0.80	Covalent bonds, HOMO-LUMO gap
10	SOLIDS	1.00■10?6	10?■■	10??	0.75	Crystalline solids, proton mass, phonons
11	LIQUIDS	1.21■10?6	10??	10?8	0.70	Water, electron density waves
12	GASES	1.44■10?6	10?8	10?7	0.65	Air molecules, ideal gas
13	PLASMA	1.69■10?6	10?7	10?6	0.60	Solar corona, Langmuir waves
14	Molecular clusters	1.96■10?6	10?6	10?5	0.55	Proteins, colloids
15	Cellular structures	2.25■10?6	10?5	10?4	0.50	Membranes, organelles
16	Macroscopic matter	2.56■10?6	10?4	10?■	0.45	Dust grains

Level	State Description	ρ_n (J/m ³)	E_n (J)	Scale (m)	η_i	Physical Examples
17	Centimeter objects	2.89×10^6	10^2	10^2	0.40	Rocks, organisms
18	Meter-scale	3.24×10^6	10^2	10^3	0.35	Buildings, trees
19	Geological (km)	3.61×10^6	10^2	10^4	0.30	Mountains, lakes
20	Planetary	4.00×10^6	1 J	10^6	0.25	Earth, Moon, Mars (Ug4 anchor)
21	Stellar	4.41×10^6	10 J	10^7	0.20	Sun, red dwarfs, white dwarfs
22	Solar system	4.84×10^6	10^8	10^{10}	0.15	Heliosphere, Kuiper belt
23	Interstellar	5.29×10^6	10^9	10^{15}	0.12	Nebulae, star clusters
24	Galactic	5.76×10^6	10^4	10^{18}	0.10	Spiral arms, galactic disk
25	Supercluster	6.25×10^6	10^5	10^{22}	0.08	Galaxy groups, Laniakea
26	Universal	6.76×10^6	10^6 J	10^{26}	0.05	Observable universe, Hubble volume

2. Universal Inertia Coupling

2.1 Definition

The Universal Inertia at level i :

$$U_{\{i, \text{level}\}} = \lambda_i \cdot \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \cdot \omega_{\text{LENR}} \cdot \cos(\pi t_n) \cdot (1 + f_{\text{TRZ}})$$

where:

η_i = level-dependent coupling constant (Table above, column η_i)

$\rho_{\text{SCm}}/\rho_{\text{UA}} = 10^8/10^{18} = 10^{-10}$ (vacuum density ratio)

$\eta_{\text{LENR}} = 1.25 \times 10^{12}$ Hz (LENR resonance frequency)

t_n = negative time parameter (cosine modulation)

f_{TRZ} = time-reversal zone factor (default 0.01)

2.2 Level-10 Reference Value

For the solid-state reference (level 10, $t_n = 0$, $f_{\text{TRZ}} = 0.01$):

$$U_{\{i=10\}} = 0.75 \times 10^3 \times 1.25 \times 10^{12} \times 1.0 \times 1.01 = 9.47 \times 10^{14} \text{ J/m}^3 \cdot \text{Hz}$$

Validator confirms: Universal Inertia Level 10 ? PASS ?

3. Core UQFF Gravity Equation

The gravitational field at position (r, t) is the 26-layer superposition:

$$\mathbf{g}(r, t) = \sum_{i=1}^{26} \left[\mathbf{U}_{g1,i}(r) + \mathbf{U}_{g2,i}(r) + \mathbf{U}_{g3,i}(r, t) + \mathbf{U}_{g4,i}(r, t) \right]$$

where each contributes a distinct physical mechanism:

Ug1_i: Magnetic dipole buoyancy (SOURCE52): $Ug1_i = (E_{DPM}/r) \cdot \rho_{UA} \cdot f_{TRZ}$

Ug2_i: Charge-reactivity (SOURCE54): $Ug2_i = s_{field} \cdot [UA]_i \cdot r$

Ug3_i: String rotation (SOURCE56): $Ug3_i = O_{string} \cdot \rho_{SCm} \cdot \sin(i\pi/26)$

Ug4_i: Vacuum concentration (SOURCE57): $Ug4_i = M_{source} \cdot \rho_{vac}/(d \cdot E_{LEP})$

4. Dual Consistency of the Two Representations

The polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{SCm} \cdot n$) representations are related through the characteristic volume V_n at each level:

$$E_n = \rho_n \cdot V_n = \rho_{SCm} \cdot n^2 \cdot V_n = 10^{(n-20)} \text{ J}$$

$$\Rightarrow V_n = \frac{10^{(n-20)}}{\rho_{SCm} \cdot n^2} = \frac{10^{(n-20)}}{10^{(n-8)}} \cdot \frac{1}{n^2} = \frac{10^{(n-12)}}{n^2} \text{ m}^3$$

This defines the **characteristic volume** at level n as the volume over which the polynomial energy is distributed at the local vacuum density. For level 10: $V_{10} = 10^4/(100) = 10^4 \text{ m}^3$ as a cube of side $\sim 0.046 \text{ m}$ (4.6 cm), consistent with the 10^4 m typical scale as 10^4 lattice sites in a mole of solid.

5. Nuclear Binding Energy Check

Level 8 provides an observable verification point. The validator reports:

$$E_8 = 10^4 \text{ J} = 6.25 \text{ MeV}$$

Expected nuclear binding per nucleon: 8 MeV

Error: 21.97% (within 50% tolerance)

Validator: Test 1 Nuclear Binding Check ? PASS ? (at 21.97% error < 50% tolerance)

This 22% discrepancy at level 8 reflects the difference between the UQFF polynomial (purely geometric/exponential) and the QCD-derived nuclear binding energy. The UQFF 26-level polynomial is an energy scale index, not a precision nuclear physics formula but it correctly locates level 8 within the nuclear binding energy decade ($10^4 \text{ J} \sim 6 \text{ MeV}$).

Conclusions

The UQFF 26-level energy hierarchy provides a self-consistent, geometrically structured energy index spanning 25 decades. Both the polynomial ($E_n = 10^{(n-20)}$) and density ($\rho_n = \rho_{SCm} \cdot n$) representations are validated. Levels 10-13 correspond to the four classical matter states (solid/liquid/gas/plasma), with level 20 anchoring the Ug_4 galactic vacuum scale and level 26 marking the observable universe boundary.

Validators: QCalc_Phase1_Validation.py Test 1 PASS ? | test_phase2_validation.py 26/27 PASS | ? = 0.0005/day | [SSq] = 0.57

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-11} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{-11} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{\{g1\}} + U_{\{g2\}} + U_{\{g3\}} + U_{\{g4\}} + U_{\{bi\}} + U_m - \sum_{i=1}^4 \lambda_i \text{ig}l[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}}] \text{igr}$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420): $\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 U_m Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imes} \text{ig}l(1 + 10^{13} \backslash, \backslash \Theta(\text{ho}_{\text{SCm}} - \text{ho}_c) \text{igr}) \text{imes} \text{ig}l(1 + A_q \cos(\Delta \omega a, t) \text{igr})$$

Symbol	Value	Description
ρ_c	10^1 kg/m^3	SCm critical superconducting density

Symbol	Value	Description
A _q	0.1	Quasi-periodic beating amplitude (10%)
Δω	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	U _g _sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β _i × U _{bi}	Expanding nebulae, stellar winds
Superconductive	U _m × (1+10 ¹³ ·f _H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.